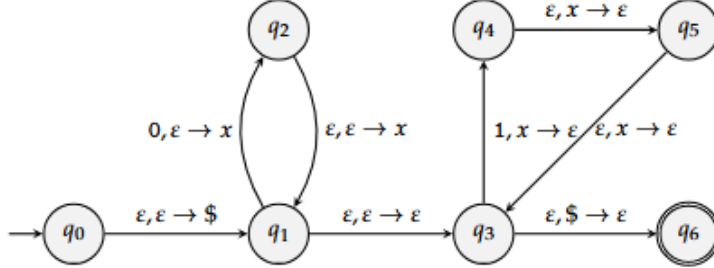


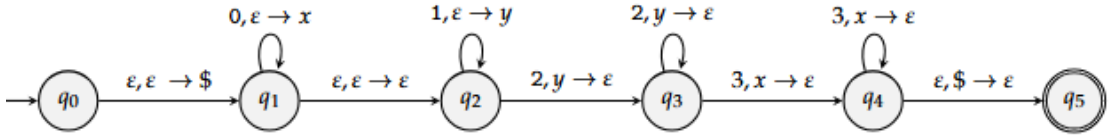
BRAC UNIVERSITY
CSE331 : Automata and Computability
Assignment 4

1. Draw the state diagram of a PDA for the following CFL:

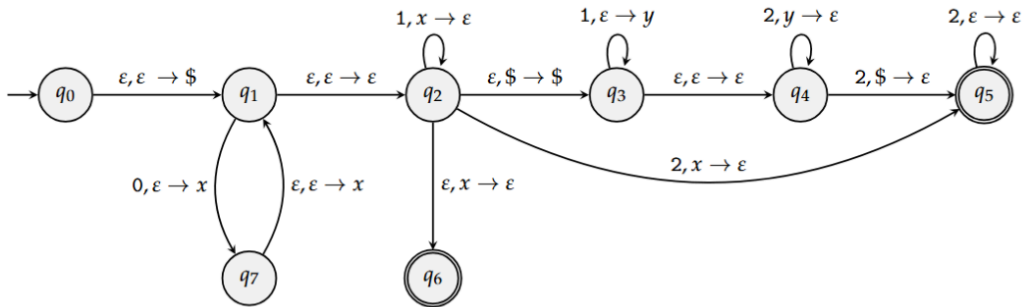
A. $L(M) \rightarrow \{0^n 1^m \mid n, m \geq 0 \text{ and } 2n = 3m\}, \text{ where } \Sigma = \{0, 1\}$



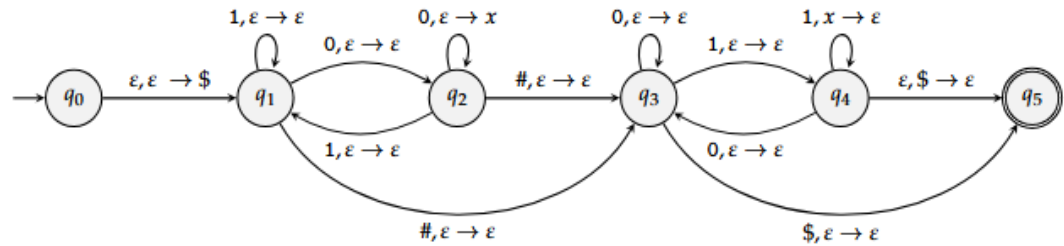
B. $L(M) \rightarrow \{0^n 1^m 2^m 3^n \mid n, m > 0\}, \text{ where } \Sigma = \{0, 1, 2, 3\}$



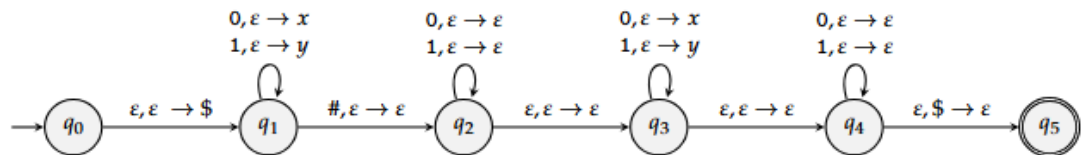
C. $L(M) \rightarrow \{w = 0^i 1^j 2^k \mid i, j, k \geq 0 \text{ and } j < 2i + k\}, \text{ where } \Sigma = \{0, 1, 2\}$



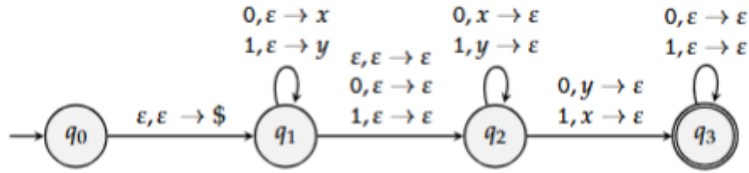
D. $L(M) \rightarrow \{w_1 \# w_2 \mid \text{the number of } 00 \text{ in } w_1 \text{ is equal to the number of } 11 \text{ in } w_2\}, \text{ where } \Sigma = \{0, 1\}$



E. $L(M) \rightarrow \{w \# x \mid w^R \text{ is a substring of } x\}, \text{ where } \Sigma = \{0, 1\}$

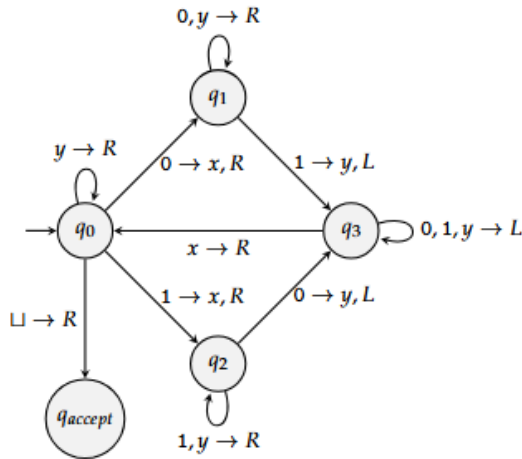


F. $L(M) \rightarrow \{w \mid w \text{ is not a palindrome}\}, \text{ where } \Sigma = \{0, 1\}.$

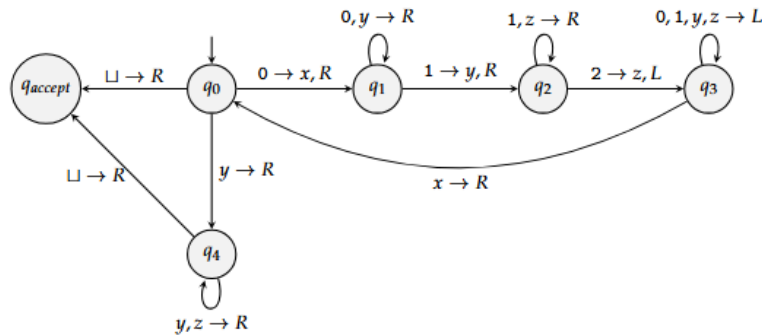


2. Draw the state diagram of a TM that decides the following languages:

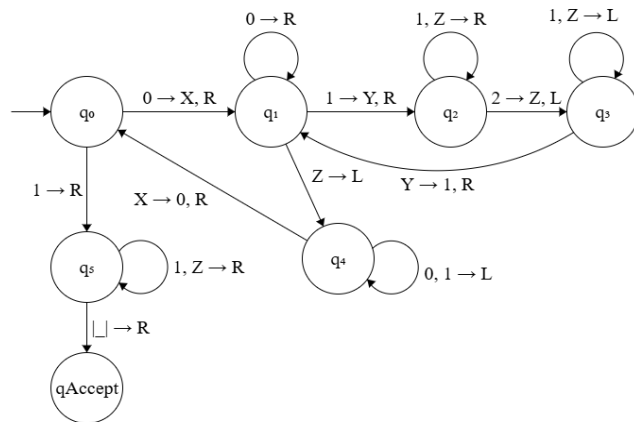
A. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains equal numbers of 0s and 1s}\}$, where $\Sigma = \{0, 1\}$



B. $L(M) \rightarrow \{0^n 1^n 2^n \mid n \geq 0\}$, where $\Sigma = \{0, 1, 2\}$



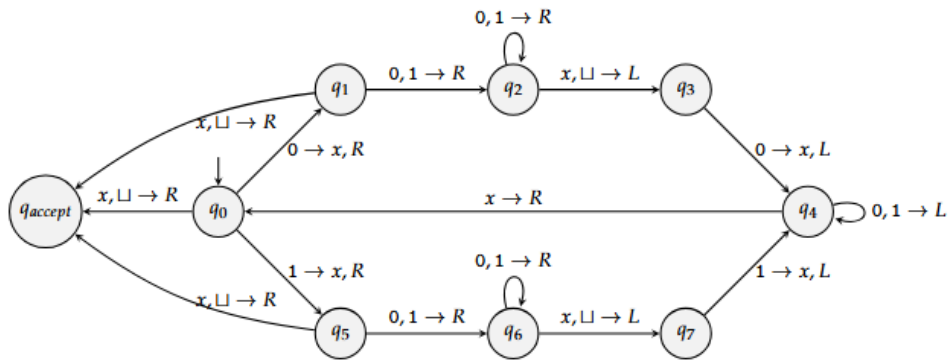
C. $L(M) \rightarrow \{0^i 1^j 2^k \mid i, j, k > 0 \text{ and } k = i * j\}$, where $\Sigma = \{0, 1, 2\}$



D. $L(M) \rightarrow \{0^{2^n} \mid n \geq 0\}$, where $\Sigma = \{0\}$

See example 3.7 from the book.

E. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ is a palindrome}\}$, where $\Sigma = \{0, 1\}$



3. Prove that the following languages are decidable.

- A. $L = \{\langle G \rangle \mid G \text{ is a connected undirected graph}\}$
See example 3.23 from the book.
- B. $L = \{\langle A, B \rangle \mid A \text{ and } B \text{ are DFAs and } L(A) = L(B)\}$
See theorem 4.5 from the book.

4. Show that the following statements are true:

- A. The collection of decidable languages is closed under Union.
See problem 3.15.a from the book.
- B. The collection of decidable languages is closed under Complement.
For any decidable languages L , let M be the TM that decides L . We construct a TM M' that decides the complement of L .
 M' = "On input w :
 1. Simulate M on w .
 2. If M rejects, accept. Otherwise, reject."
Since M is a decider, M' must be a decider.

5. Given $A_{TM} \rightarrow \{\langle M, w \rangle \mid M \text{ is a TM that accepts } w\}$ is undecidable, prove that the following languages are undecidable.

- A. $RL_{TM} \rightarrow \{\langle M, w \rangle \mid M \text{ is a TM such that } L(M) \text{ is a regular language}\}$

See theorem 5.3 from the book.

- B. $M_{000} \rightarrow \{\langle M \rangle \mid M \text{ is a TM that accepts } 000\}$

Assume that the TM M decides M_{000} and let A be the TM that decides A_{TM}

A = "On input $\langle M, w \rangle$:

1. Construct the following machine M_1 :
 M_1 = "On input x :
 1. If $x \neq 000$, reject.
 2. If $x = 000$, run M on w and accept if M accepts; otherwise reject."
2. Run M on $\langle M_1 \rangle$.
3. If M accepts, accept; otherwise reject."

As M is a decider, A is also a decider. This forms a contradiction.

Hence, as A_{TM} is undecidable, M_{000} cannot be decided.